

Monopile Installation Record HB-A09

Harrow Bay Offshore Wind · Foundations

Elena Márquez · 01 July 2026 · HB-CON-IR-A09

Position	Date	Vessel	Contract
HB-A09	14 Jun 2026	MV Sea Atlas	WE-CON-2025-040 / -027

1. Position and pile data

Item	Design	As-installed
Coordinates (WGS84)	53.92880 N, 1.16540 E	within 24 cm of target
Water depth (m LAT)	33.0	33.1
Pile length (m)	77.0	77.0
Pile mass (t)	1,343	–
Target penetration (m)	34.1	34.4
Verticality	≤ 0.25°	0.2°

2. Driving record

Item	Value
Start of soft start	09:36
Driving duration	4 h 02 min
Total blow count	5,034
Maximum hammer energy	2,878 kJ
Self-weight penetration	4.0 m
Interruptions	None

3. Noise and marine mammal mitigation

Item	Value
Bubble curtain	Double big bubble curtain, both rings operational
Peak SEL at 750 m	157.4 dB (limit 160 dB)
Pre-piling search (MMO + PAM)	30 min, marine mammal detected – start delayed 15 min
Acoustic deterrent device	Activated 30 min before soft start

4. Transition piece

Item	Value
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TP installed	30 Jun 2026
Flange flatness after installation	1.0 mm (limit 2.0 mm)
Bolt tensioning	148 × M72 – completed
TP level	+0.04° (limit 0.25°)

5. Non-conformances

NCR-HB-078: minor coating damage on pile top during upending, repaired offshore.

6. Sign-off

Role	Name	Date
Offshore Installation Supervisor	Atlas Marine Contractors	15 Jun 2026
Client Representative	Felix Brandt	15 Jun 2026
Reviewed	Elena Márquez	16 Jun 2026